

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.05, beta.var.scale = 1.25\n')
```

```
## sdev = 0.05, beta.var.scale = 1.25
```

Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.2150  0.2422  0.2500  0.2496  0.2572  0.2794

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5744  0.7072  0.6557  0.7961  0.9600
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5617  0.6911  0.6398  0.7756  0.9250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.01778 0.04500 0.07111 0.23747 0.26556 1.00000
```

```

Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02389 0.04556 0.06722 0.20834 0.20722 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01889 0.05222 0.13444 0.33895 0.63944 1.00000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.03833 0.05222 0.05611 0.05651 0.06056 0.07667

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.4094 0.5667 0.5209 0.6772 0.9317
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.4011 0.5522 0.5043 0.6554 0.8611
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01833 0.44778 0.50478 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.008333 0.046111 0.331912 0.869861 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.6128 1.0000 0.7687 1.0000 1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.005556 0.013333 0.015556 0.016077 0.018889 0.031667

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2789  0.4544  0.4168  0.5794  0.9000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2811  0.4506  0.4085  0.5694  0.8194
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.9783  1.0000  0.8421  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.001111 0.011944 0.293756 0.766111 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.5967  1.0000  0.7663  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.001667 0.008333 0.011111 0.011796 0.014444 0.031667

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1611  0.3478  0.3285  0.4872  0.8900
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1683	0.3511	0.3236	0.4822	0.7683

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9039	1.0000	0.8195	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001389	0.309937	0.995833	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8892	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.003333	0.010000	0.012222	0.013519	0.016111	0.073333

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0650	0.2450	0.2460	0.3867	0.8544

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.06889	0.25111	0.24281	0.38444	0.74056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1561	0.8428	0.6264	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2809	0.8456	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8632	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001667	0.006667	0.008333	0.008999	0.010556	0.046667

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.06444	0.24917	0.24721	0.38667	0.83944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0700	0.2544	0.2436	0.3856	0.7017

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.7267	0.5699	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2695	0.7121	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9961	1.0000	0.8481	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0055556	0.0072222	0.0076178	0.0088889	0.0288889

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.00000 0.02389 0.18722 0.19984 0.32278 0.79389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02667	0.19611	0.19801	0.32556	0.66222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3272	0.4583	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4311	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1982	0.0000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2828	0.9739	0.6806	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0038889	0.0050000	0.0055529	0.0066667	0.0188889

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01611	0.16694	0.18298	0.29889	0.77944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01889 0.17667 0.18311 0.30556 0.65222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.06222 0.38780 0.94444 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 1.00000 1.00000 0.9985 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.1297 0.0000 1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.1967 1.0000 0.6813 1.0000 1.0000

--- Summary for K = 9 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.002222 0.006667 0.008889 0.008996 0.011111 0.023333

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003889 0.116111 0.147030 0.246111 0.725556
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.125556 0.145850 0.253333 0.580556
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.2503 0.5450 1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 1.00000 1.00000 0.9755 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.08889 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```
0.0000 0.0000 0.3067 0.4232 0.9311 1.0000
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2150 0.2422 0.2500 0.2496 0.2572 0.2794
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5744 0.7072 0.6557 0.7961 0.9600
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5617 0.6911 0.6398 0.7756 0.9250
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01778 0.04500 0.07111 0.23747 0.26556 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02389 0.04556 0.06722 0.20834 0.20722 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01889 0.05222 0.13444 0.33895 0.63944 1.00000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03833 0.05222 0.05611 0.05651 0.06056 0.07667
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.0000 0.4094 0.5667 0.5209 0.6772 0.9317
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.4011 0.5522 0.5043 0.6554 0.8611
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.01833 0.44778 0.50478 1.00000 1.00000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.008333 0.046111 0.331912 0.869861 1.000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.6128 1.0000 0.7687 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.005556 0.013333 0.015556 0.016077 0.018889 0.031667
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.2789 0.4544 0.4168 0.5794 0.9000
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.2811 0.4506 0.4085 0.5694 0.8194
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9783 1.0000 0.8421 1.0000 1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.001111 0.011944 0.293756 0.766111 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5967 1.0000 0.7663 1.0000 1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001667 0.008333 0.011111 0.011796 0.014444 0.031667
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1611 0.3478 0.3285 0.4872 0.8900
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1683 0.3511 0.3236 0.4822 0.7683
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9039 1.0000 0.8195 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.001389 0.309937 0.995833 1.000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 1.0000 1.0000 0.8892 1.0000 1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.003333 0.010000 0.012222 0.013519 0.016111 0.073333
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0650 0.2450 0.2460 0.3867 0.8544
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.06889 0.25111 0.24281 0.38444 0.74056
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1561 0.8428 0.6264 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2809 0.8456 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 1.0000 1.0000 0.8632 1.0000 1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001667 0.006667 0.008333 0.008999 0.010556 0.046667
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.06444 0.24917 0.24721 0.38667 0.83944
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0700 0.2544 0.2436 0.3856 0.7017
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.7267 0.5699 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2695 0.7121 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9961 1.0000 0.8481 1.0000 1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.0005556 0.0055556 0.0072222 0.0076178 0.0088889 0.0288889
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02389 0.18722 0.19984 0.32278 0.79389
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02667 0.19611 0.19801 0.32556 0.66222
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.3272 0.4583 1.0000 1.0000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4311  1.0000  1.0000  0.9999  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1982  0.0000  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.2828  0.9739  0.6806  1.0000  1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min.  1st Qu.    Median      Mean   3rd Qu.     Max.
## 0.0005556 0.0038889 0.0050000 0.0055529 0.0066667 0.0188889
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01611 0.16694 0.18298 0.29889 0.77944
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01889 0.17667 0.18311 0.30556 0.65222
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.06222 0.38780 0.94444 1.00000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.9985  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1297  0.0000  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.1967  1.0000  0.6813  1.0000  1.0000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min.  1st Qu.    Median      Mean   3rd Qu.     Max.
## 0.002222 0.006667 0.008889 0.008996 0.011111 0.023333
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.003889 0.116111 0.147030 0.246111 0.725556
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.125556 0.145850 0.253333 0.580556
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2503  0.5450  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.9755  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.08889 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.3067  0.4232  0.9311  1.0000
##
## ‘‘‘

```

Conclusion

and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.